Appendix 1: Trout Lake

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Overview

Here we show the LM, and GAM analysis for the Trout Lake dataset. For complete code, see the full quarto document

Approach for running LTER-NTL Trout Lake data using temporally structured GAMs and linear regression models with time periods defined a-priori.

First we start with looking at three time periods using linear regression models:

1. The historical regime with low water clarity
2. A clear water regime where the introduction of Lake Trout into the system from stocking in 2006.
3. A novel regime following the introduction of invasive, predatory water flea (*Bythotrephes*) in 2014 which lead to a reversion of water clarity to a less clear state.

To examine this we start by looking at a few key food web conditions using intercept only models through time:

1. Water clarity
2. Phosphorus - which impacts water clarity and is a common bottom-up process that could impact water clarity and we examine as an alternative hypothesis to the top down processes of Lake Trout and invasive speceis.
3. Abundance of large zooplankton *Daphnia* and *Calanoids*
4. Chlorophyll

LM

We approach this by fitting a linear model with the a priori time periods as a factor and compare AIC to a single intercept model for each variable.

We find that water clarity, chlorophyll, and large zooplankton abundance are all better explained by a model that includes *a priori* defined time periods improves model fit (Table S1). This indictes that the means for these variables change through time. By examining the time series plot (Figure S1) we can see that large zooplankton and water clarity appear more correlated after 2006, when Lake Trout were stocked in the system. This aligns with our hypothesis that the introduction of top down control in the system has made large zooplankton abundance more tightly coupled with water clarity.

Next we consider whether there are changing relationships between water clarity and these ecosystem dynamics by including a slope parameter and its interaction with time period. We only include an investigation into large zooplankton-water clarity relationship in the main text, but here we also explore chlorophyll-water clarity.

Both chlorophyll and large zooplankton best explain water clarity with a time-varying relationship model, rather than a time-varying abundance model. We can look at the model residuals to see how they compare for a time-varying model versus a time-invariant model. Using the large zooplankton covariate as an example, we can see in the timeseries of residuals (Figure S2) that there is a temporal trend, supporting a temporally structured model. We find the q-q plot is similar across the models (Figure S3) and are mostly okay.

Next, we examine how these relationships change through time, with a focus on large zooplankton.

We can see from these results that there was a weak relationship from 1980 - 2006 between zooplankton abundance and water clarity, that switched to a strong, positive relationship in 2007 - 2014 as large zooplankton were released from top down pressure, and a slightly stronger relationship after 2014 with the introduction of *Bythotrephes* (Figure S4 & Table S3).

GAM

If we take the Zooplankton-water quality relationship and fit a temporally structured GAM and compare it to the linear model, we find similar results. Both predictions also acccurately identify 2006 as a breakpoint and a second break point in 2016, although the GAM indicates that the slope of this relationship is a continuously changing trend (Figure S8) and the intercept more closely tracks abundance (Figure S9).

We can do similar residual diagnostics with the GAM model used. We can see that in a model that does not include a smooth by year term, there is a temporal trend in the residuals, but this is resolved when a smooth by year is implemented (Figure S6). There doesn't appear to be issues with the QQ plots for any specification (Figure S7).

Figures & Tables

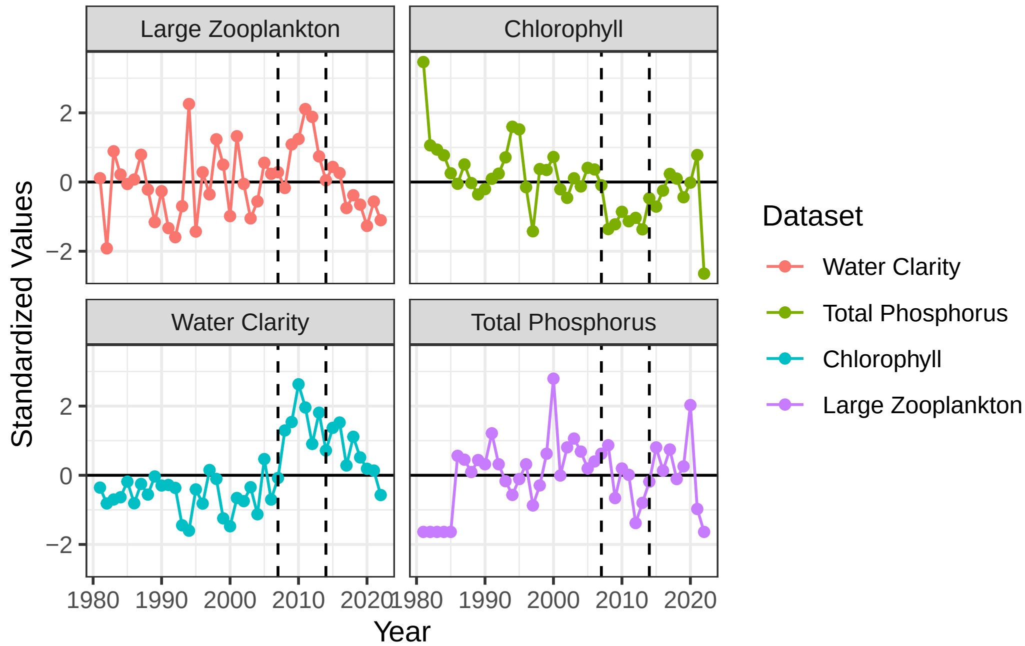

Figure S1: Standardized time series of ecological conditions observed in Trout Lake.

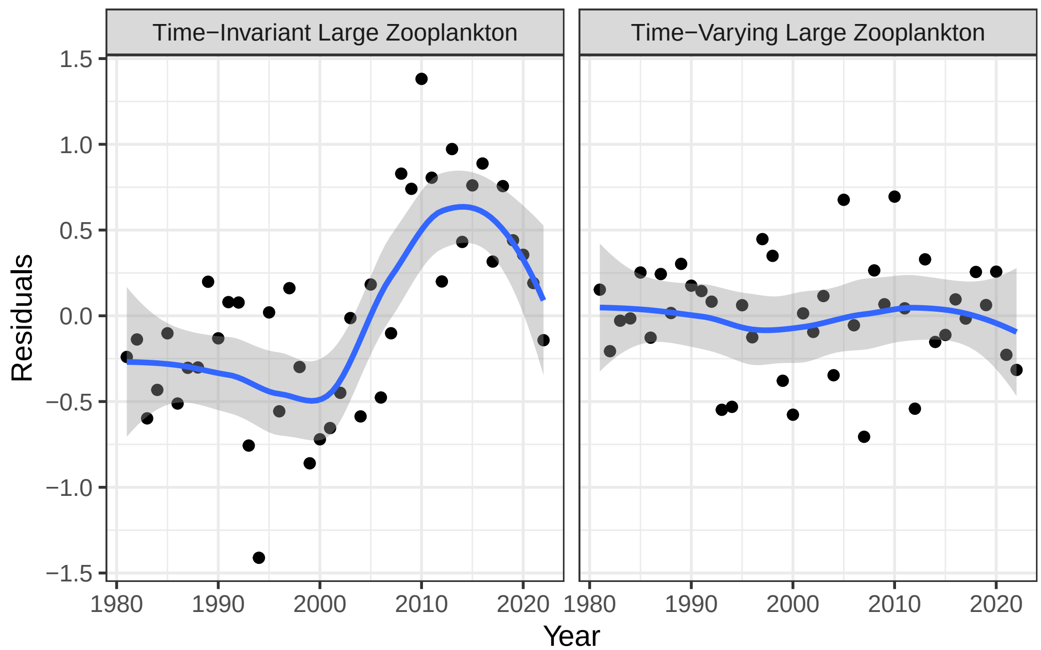

Figure S2: Model residuals for the time-varying water clarity - zooplankton model compared to a time-invariant model where large zooplankton is used as a covariate without *a priori* defined time peariods

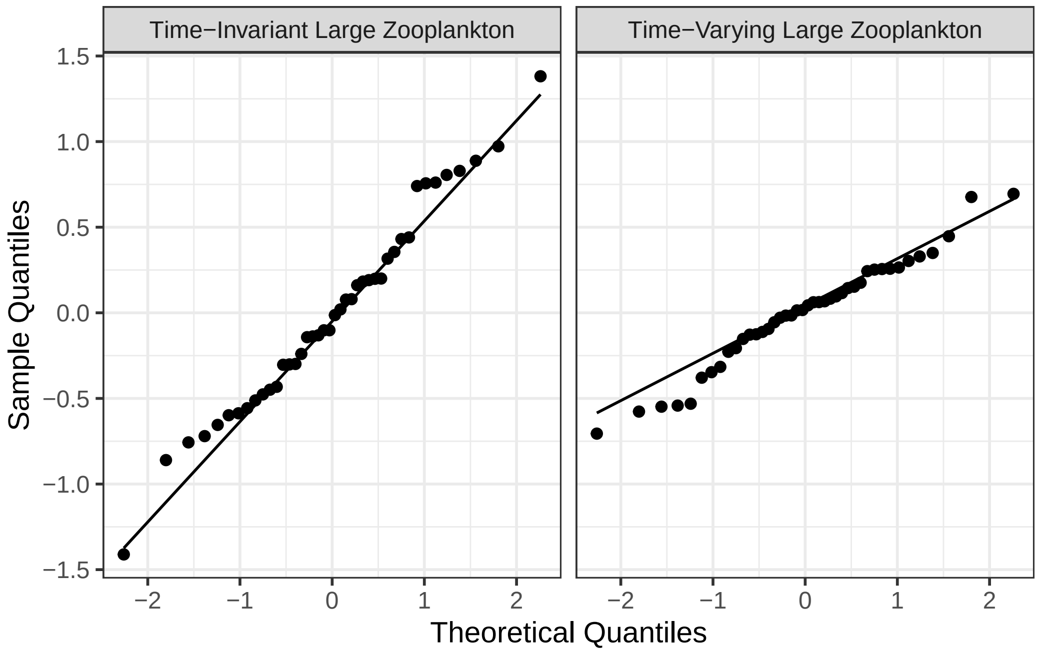

Figure S3: Q-Q plot for the time-varying water clarity - zooplankton model compared to a time-invariant model where large zooplankton is used as a covariate without *a priori* defined time peariods

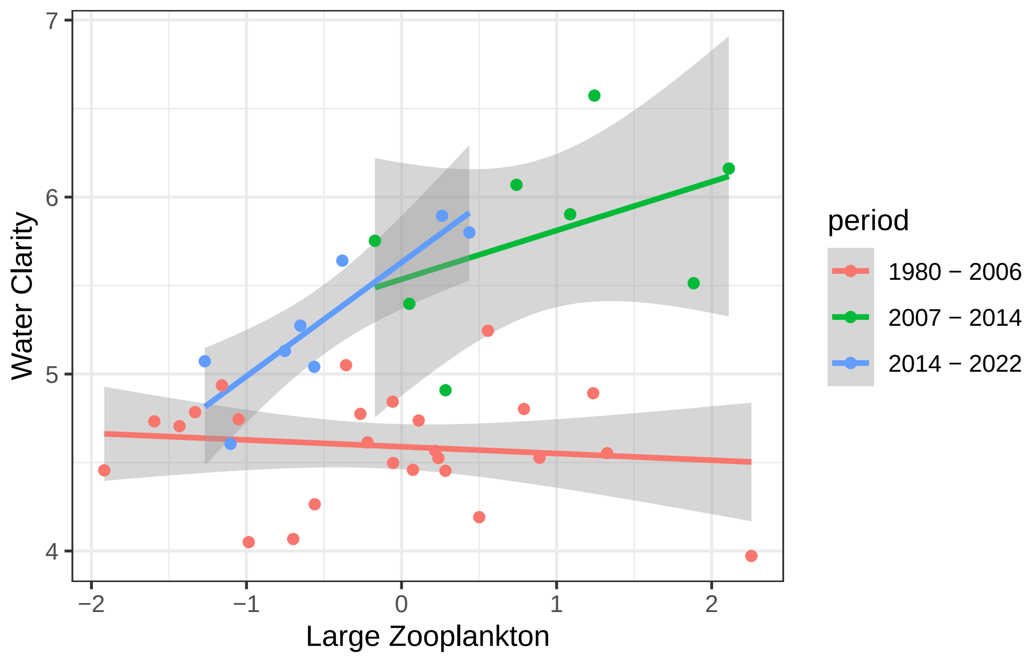

Figure S4: Relationship between large zooplankton abundance and water clarity for three time periods defined *a priori*.

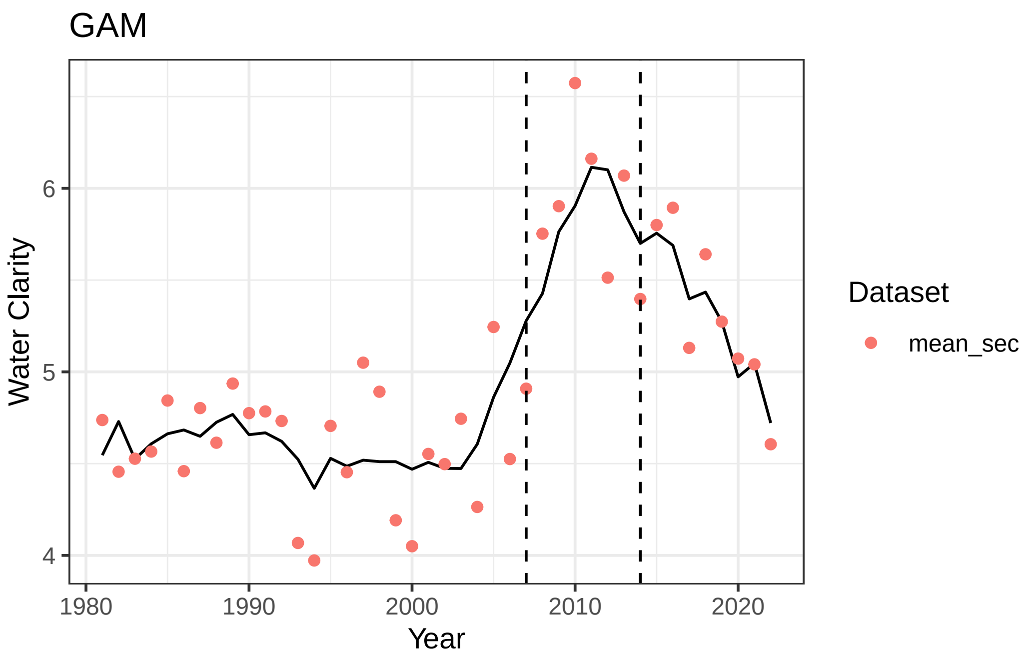

Figure S5: Predicted values of water clarity (black line) compared to observed values (red points) using a time-varying relationship with zooplankton abundance.

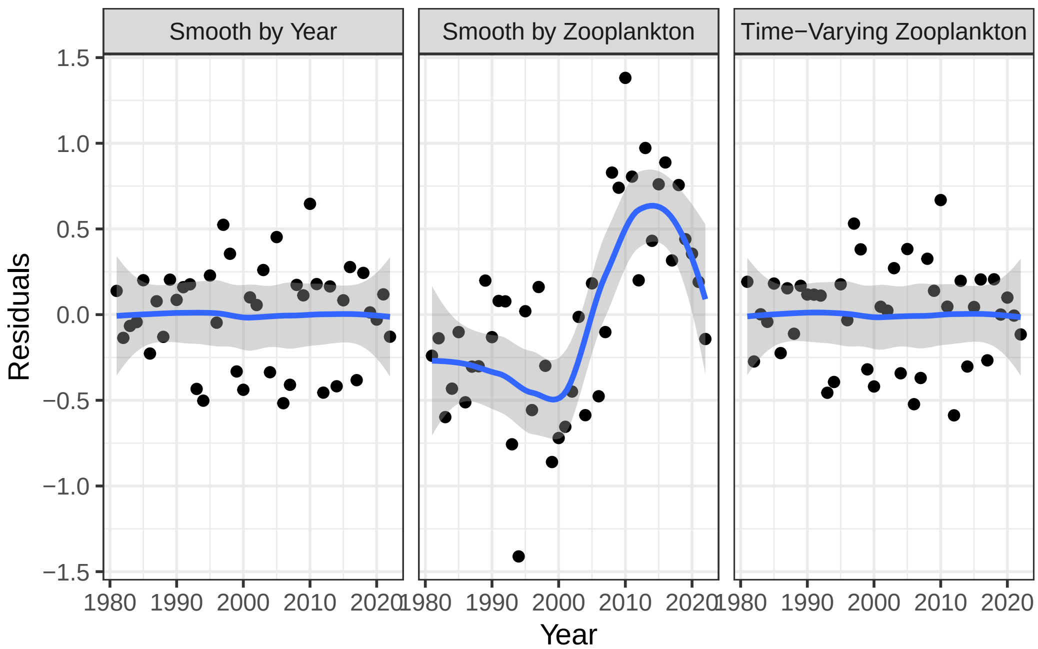

Figure S6: Model residuals for a smooth on zooplankton abundance, a smooth on year, and a smooth on year by zooplankton.

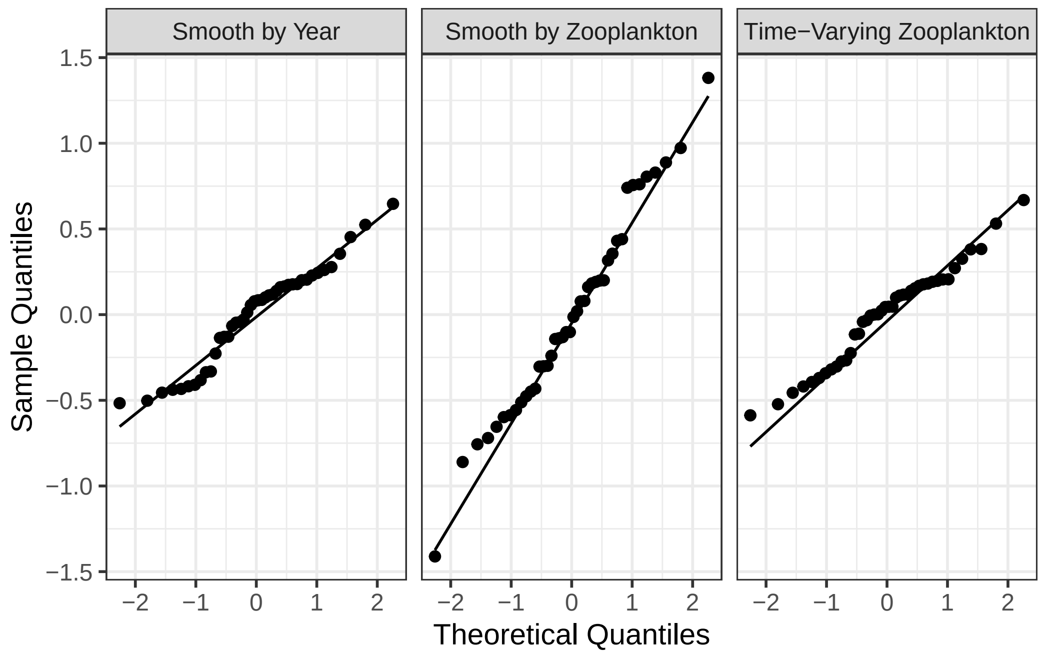

Figure S7: Q-Q plot for a smooth on zooplankton abundance, a smooth on year, and a smooth on year by zooplankton.

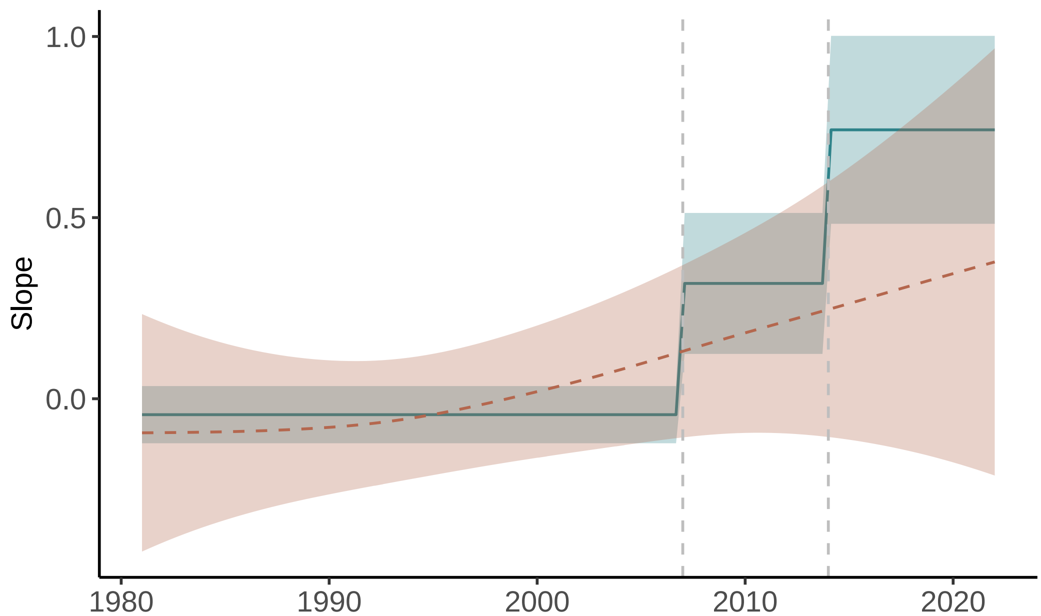

Figure S8: Estimates of the slope of the relationship based on a time-varying LM (blue) and a time-varying GAM (red)

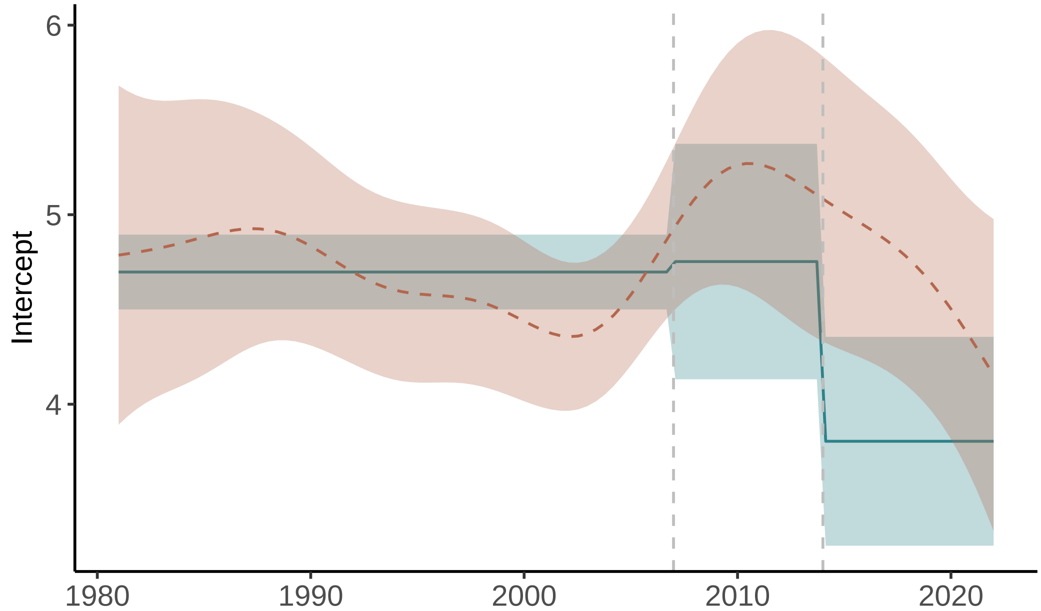

Figure S9: Estimates of the intercept of the relationship based on a time-varying LM (blue) and a time-varying GAM (red)

Variable	No.Period.AIC	Period.AIC	Best.Model
Water Clarity	122.18	83.43	Period
Total Phosphorus	122.18	125.75	No difference
Chlorophyll	122.18	111.00	Period
Large Zooplankton	122.18	115.84	Period

Table S1: AIC values comparing linear models with *a priori* defined time periods (time-varying model) and no time period (time-invariant model)

Covariates	Period	Interaction
Chlorophyll	43.99	39.56
Total Phosphorus	44.45	46.31
Large Zooplankton	43.58	35.99

Table S2: AIC values comparing linear models comparing time-varying abundance models versus time-varying relationship models

term	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	4.70	0.20	23.80	0.00
Large	-0.04	0.08	-0.56	0.58
as.factor(period)2	0.06	0.62	0.09	0.93
as.factor(period)3	-0.89	0.55	-1.62	0.11
Large:as.factor(period)2	0.36	0.19	1.86	0.07
Large:as.factor(period)3	0.79	0.26	3.03	0.00

Table S3: Model coefficients for the time-varying relationship model between large zooplankton and water clarity

Appendix 2: Lake Washington Analysis

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Here we show the GAM and DLM analysis for the Lake Washington dataset. For complete code, see the full quarto document. We include some code snippets here for reference.

Data Processing

In Lake WA the major change ecosystem change is related to sewage in the lake – adding wastewater plants in the 1960s reduced phosphorous (Figure S1) and in turn, blue green algae (Figure S2) which can be seen by examining time series plots of the data.

We can then use the MARSS package to fit a DLM. We'll filter the data to only use counts before 1974, and log transform densities. We also z-score the covariate. This is a good example of imperfect data – there's a couple of missing observations in the covariate, which we can't have in a DLM or GAM – so let's just fill them in with a spline. We can do this as follows:

```
dat <- dplyr::filter(dat, Year < 1974) %>%
  dplyr::mutate(y = log(Bluegreens), zp = as.numeric(scale(TP)))

interpolated <- spline(dat$zp, xout = 1:nrow(dat))$y
dat$zp[which(is.na(dat$zp))] <- interpolated[which(is.na(dat$zp))]
```

The response data exhibit a strong seasonal cycle – and it's a good idea to de-season that to focus on the phosphorous effect. Here we de-season the data:

```
# Fill in a few missing values
dat$y_interp <- dat$y
interpolated <- spline(dat$y, xout = 1:nrow(dat))$y
missing <- which(is.na(dat$y_interp))
dat$y_interp[missing] <- interpolated[missing]
```

```
dat <- dplyr::group_by(dat, Month) %>%
  dplyr::mutate(month_mean = mean(y,na.rm=T)) %>%
  dplyr::ungroup() %>%
  dplyr::mutate(y_adj = y_interp - month_mean)
```

Last, we should look at the lags between driver and response relationships which we can do using a CCF (Figure S3). The CCF here indicates a 2-4 month lag, where TP and algae are more strongly associated with one another.

We'll use a lag of 4 months, which we can apply as follows:

```
lag_n <- 5
dat$lagged_zp <- c(rep(NA, lag_n), dat$zp[1:(nrow(dat)-lag_n)])
dat <- dat[-c(1:lag_n),]
```

DLMs

First, we need to define the model – this block is basically copied from the MARSS book, the salmon survival case study

```
m <- 2
TT <- nrow(dat)
B <- diag(m) ## 2x2; Identity
U <- matrix(0, nrow = m, ncol = 1) ## 2x1; both elements = 0
Q <- matrix(list(0), m, m) ## 2x2; all 0 for now
diag(Q)[1] <- 0.0000001
diag(Q)[2] <- c("q.beta")
#diag(Q) <- c("q.alpha", "q.beta") ## 2x2; diag = (q1,q2)
Z <- array(NA, c(1, m, TT)) ## NxMxT; empty for now
Z[1, 1, ] <- rep(1, TT) ## Nx1; 1's for intercept
Z[1, 2, ] <- dat$lagged_zp ## Nx1; predictor variable
A <- matrix(0) ## 1x1; scalar = 0
R <- matrix("r") ## 1x1; scalar = r
## only need starting values for regr parameters
inits_list <- list(x0 = matrix(c(0, 0), nrow = m))
## list of model matrices & vectors
mod_list <- list(B = B, U = U, Q = Q, Z = Z, A = A, R = R)
# convert response to matrix
dat_mat <- matrix(dat$y_adj, nrow = 1)
# fit the model -- crank up the maxit to ensure convergence
```

```
dlm_1 <- MARSS(dat_mat, inits = inits_list, model = mod_list,
control = list(maxit=4000), method="TMB")
```

```
MARSS fit is
Estimation method: TMB
Estimation converged in 18 iterations.
Log-likelihood: -155.082
AIC: 318.1641   AICc: 318.4626

         Estimate
R.r         0.226
Q.q.beta    0.235
x0.X1       0.498
x0.X2       1.422
Initial states (x0) defined at t=0

Standard errors have not been calculated.
Use MARSSparamCIs to compute CIs and bias estimates.
```

Let's put the state estimates back into the original dataframe and do some plotting. The slope is generally positive, which is what we expect for the hypothesized relationship between bluegreen algae and total phosphorus – and exhibits clear variation through time (Figure S4). We can also plot the fitted values from our model (Figure S5).

And we can examine the residuals for a temporal trend, and see that because the DLM is flexible, there's not much evidence of a trend (Figure S6) and there are no issues wiht the q-q plot (Figure S7).

GAMs

We can fit a GAM model 5 ways, 1) so that the smoother is a function of phosphorus which represents a nonlinear functional relationship between total phosphorus and bluegreen algae that is not temporally structured thus the relationship is stationary and fixed through time. This is how GAMs are typically structured for ecological analyses. 2) a temporally structured model such that the abundance of bluegreen algae varies as a function of time unrelated to phsophorus, 3) A temporally structured model so that the smooth function of phosphorus varies by time, and 4) a smooth on date as a driver of cyanobacteria where the smooth through time varies by phosphorus level. Finally, these relationships can be fit as 5) a tensor product which provide a multivariate function spanning the dimensions of time and phosphorus. We will not discuss tensor products in greater detail and only include them for completeness.

We show the model predicted cyanobacteria trend through time using these five modeling approaches (Figure S8). While they are quite similar, the predictions diverge sustantially at the end of the time series.

```
dat$n_date <- as.numeric(as.factor(dat$date))

gam_cr1 <- gam(y_adj ~ s(lagged_zp, bs = "cr"), data = dat) #1
gam_cr2 <- gam(y_adj ~ s(lagged_zp, by = n_date, bs = "cr"), data = dat) #2
gam_cr3 <- gam(y_adj ~ s(n_date, by = lagged_zp, bs = "cr"), data = dat) #3
gam_cr4 <- gam(y_adj ~ s(n_date), data = dat) #4
gam_cr5 <- gam(y_adj ~ te(lagged_zp, n_date), data = dat) #5
```

```
[1] "#26172AFF" "#414081FF" "#357BA2FF" "#3DB4ADFF" "#A8E1BCFF"
```

It is important to also examine the residuals from these models. For example, we can look at the residuals for the GAMs fit to the Lake Washington data. In looking across models, those that do not include time-varying parameters show evidence of a persistent trend in the residuals that is not accounted for by the model ((Figure S9) despite teh q-q plot being reasonable for all model specifications ((Figure S10).

```
resid_df_1 <- data.frame(date = dat$date, resid = resid(gam_cr1),
                         Model = "s(Phos)")
resid_df_2 <- data.frame(date = dat$date, resid = resid(gam_cr2),
                         Model = "s(Phos, by = date)")
resid_df_3 <- data.frame(date = dat$date, resid = resid(gam_cr3),
                         Model = "s(date, by=Phos)")
resid_df_4 <- data.frame(date = dat$date, resid = resid(gam_cr4),
                         Model = "s(date)")
resid_df_5 <- data.frame(date = dat$date, resid = resid(gam_cr5),
                         Model = "te(Phos, date")
resid <- rbind(resid_df_1, resid_df_2, resid_df_3, resid_df_4, resid_df_5)
gam_resid<-ggplot(resid, aes(date, resid, group = Model)) +
  geom_point() +
  geom_smooth() + theme_classic() +
  facet_wrap(~ Model) + xlab("Date") +
  ylab("Residuals") + ggtitle("GAM Residuals")
```

There are a few more analysis approaches for running GAMs including the choice of smoother function (Figure S8 & Figure S11) and the number of knots, or the "wiggliness" (Figure S12) of the smoothed relationship. We can plot these different results and include the results of the DLM for comparison, note that 'cr' and 'tp' are so similar they are difficult to discern from the plot. We can do a similar comparison for the choice of k. Here we compare a four different numbers of knots (Figure S12), we show an intentionally wide range of k (30 - 120) to illustrate the effects of k on estimated uncertainty, in practice k should be specified based on degrees of freedom and expectations of variability through time.

Figures

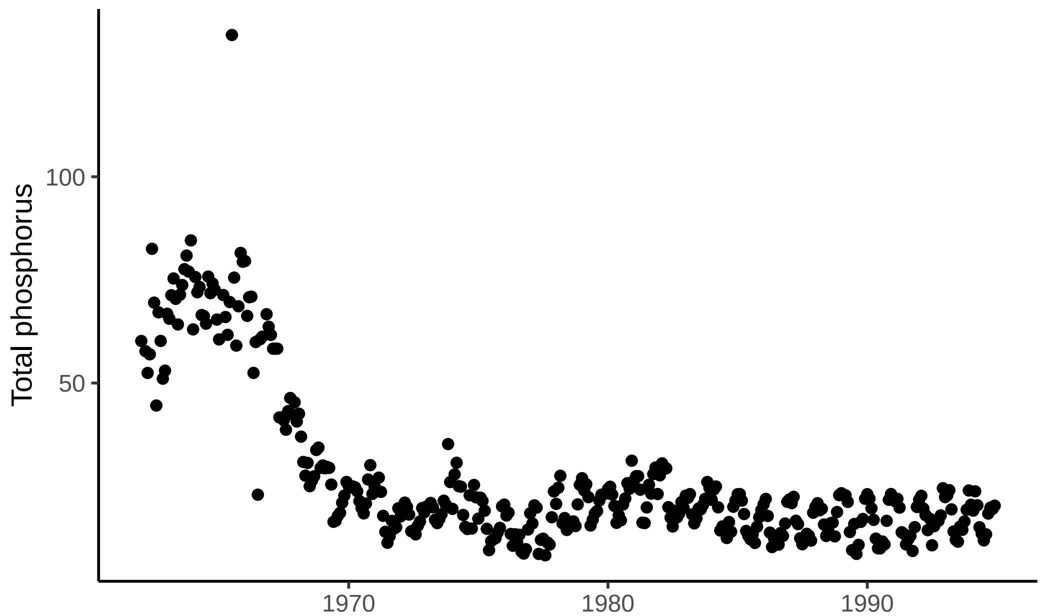

Figure S1: Time series of total phosphorus in Lake Washington

Figure S2: Time series of bluegreen algae (cyanobacteria) abundance in Lake Washington

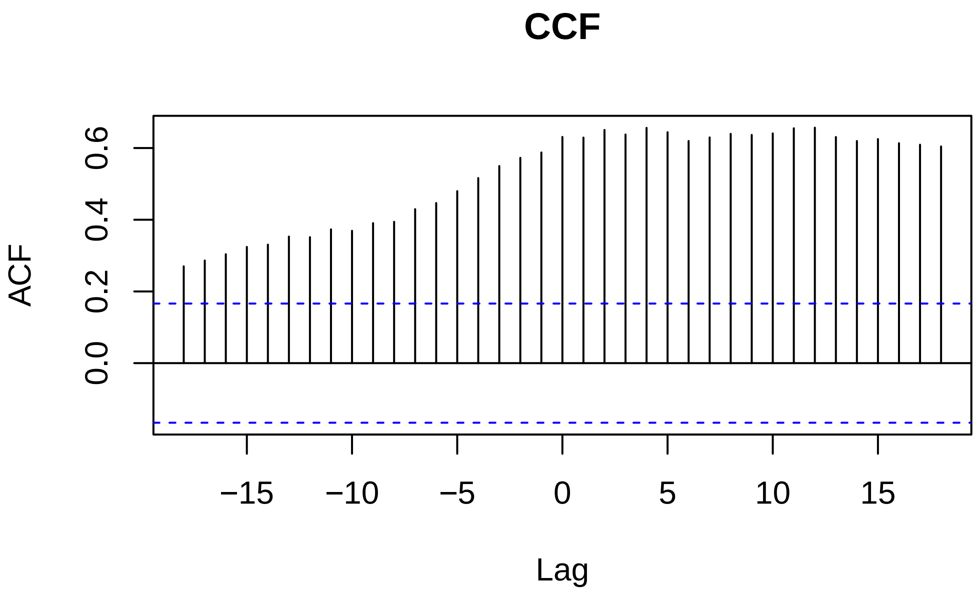

Figure S3: A CCF (Cross-Correlation Function) plot that represents the correlation between total phosphorus and blugreen algae abundance as a function of the lag (time difference) between them.

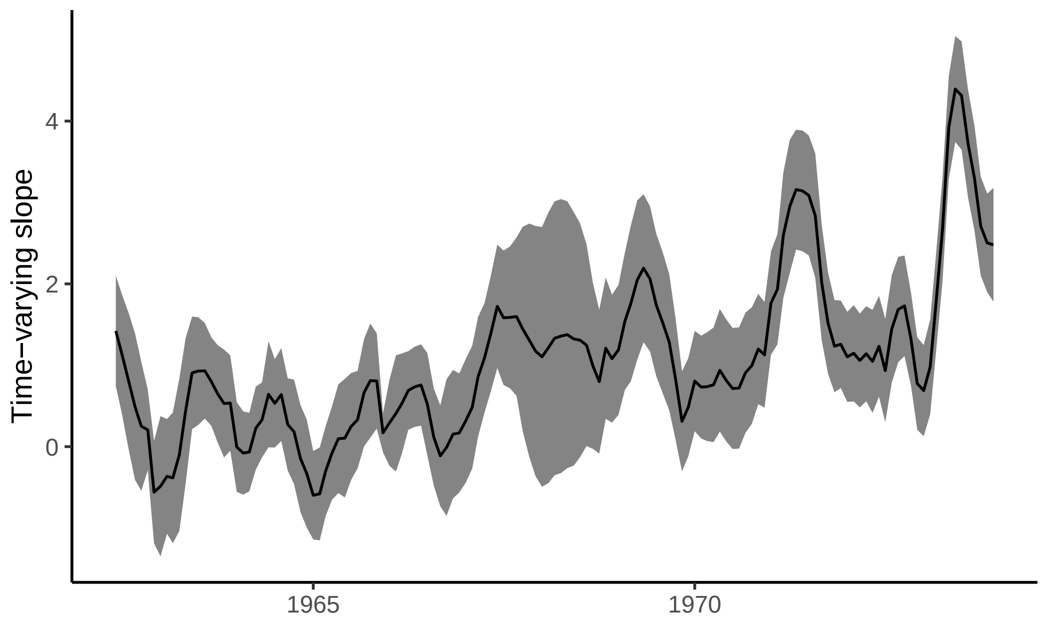

Figure S4: Time-varying slope of the relationship between bluegreen algae (response) and total phosphorus (driver) from a DLM

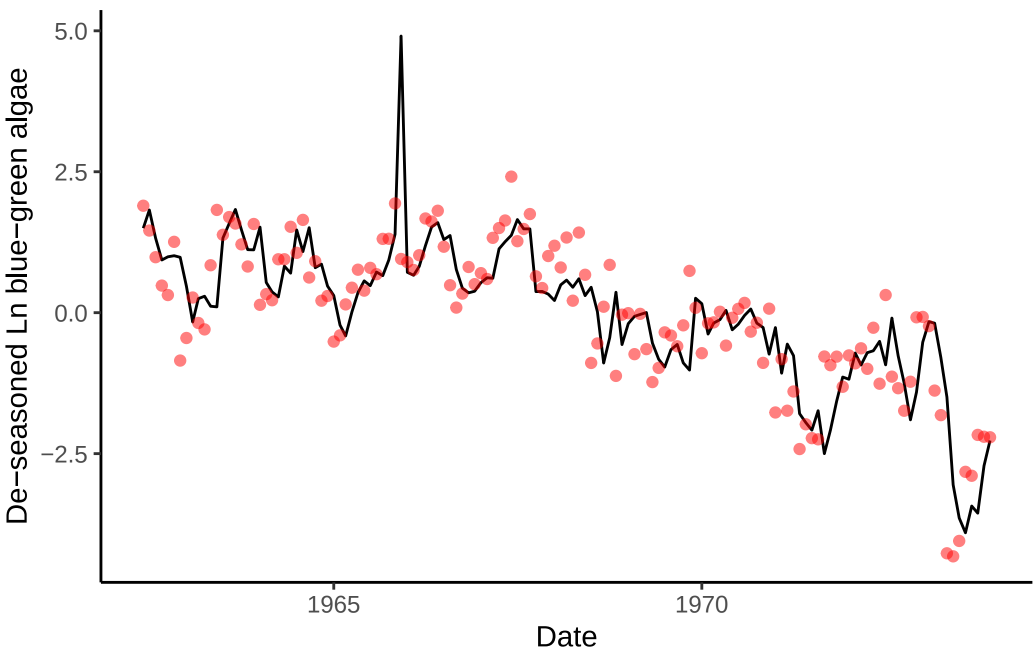

Figure S5: Fit of the DLM model with time-varying slope (black line) to observed bluegreen algae abundance (red points).

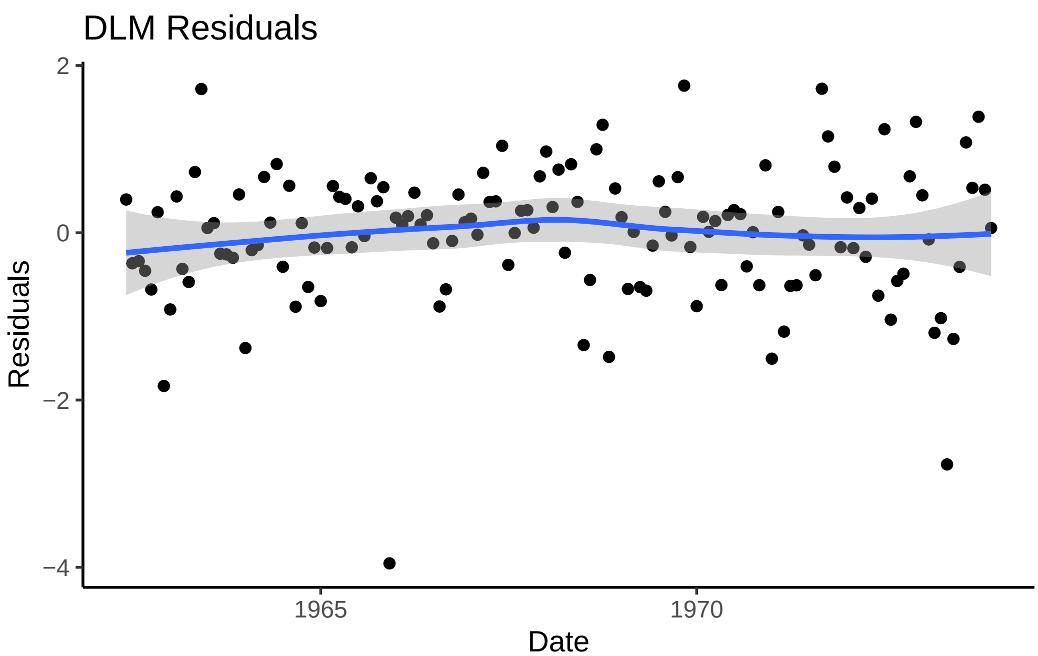

Figure S6: Time series of residuals from the DLM model fit with a time-varying slope between blue-green algae and phosphorus

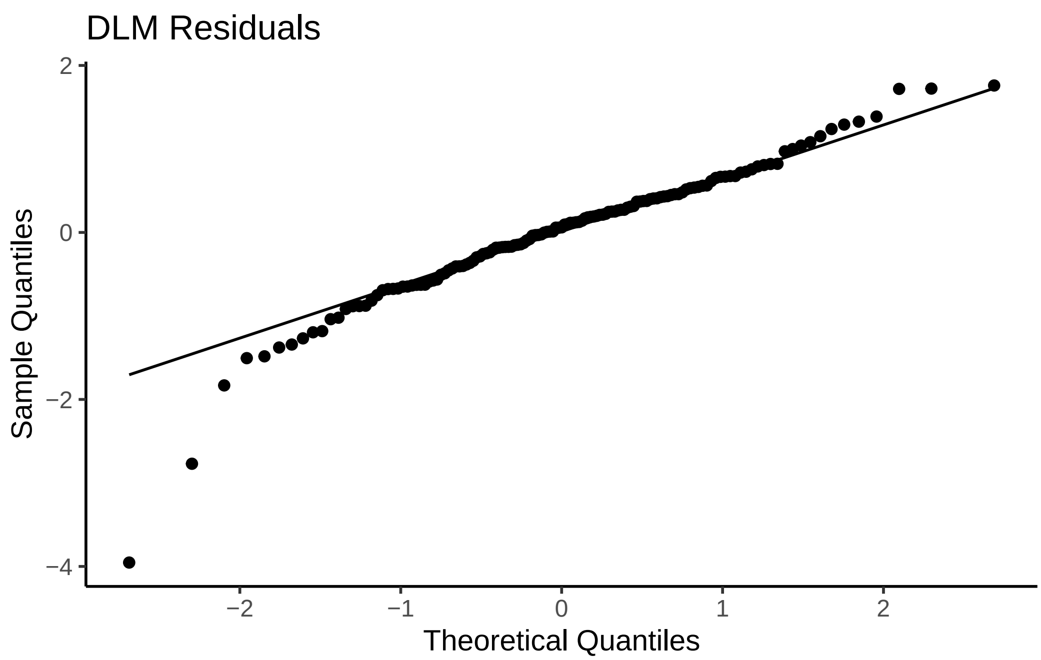

Figure S7: Q-Q plot from the DLM model fit with a time-varying slope between blue-green algae and phosphorus

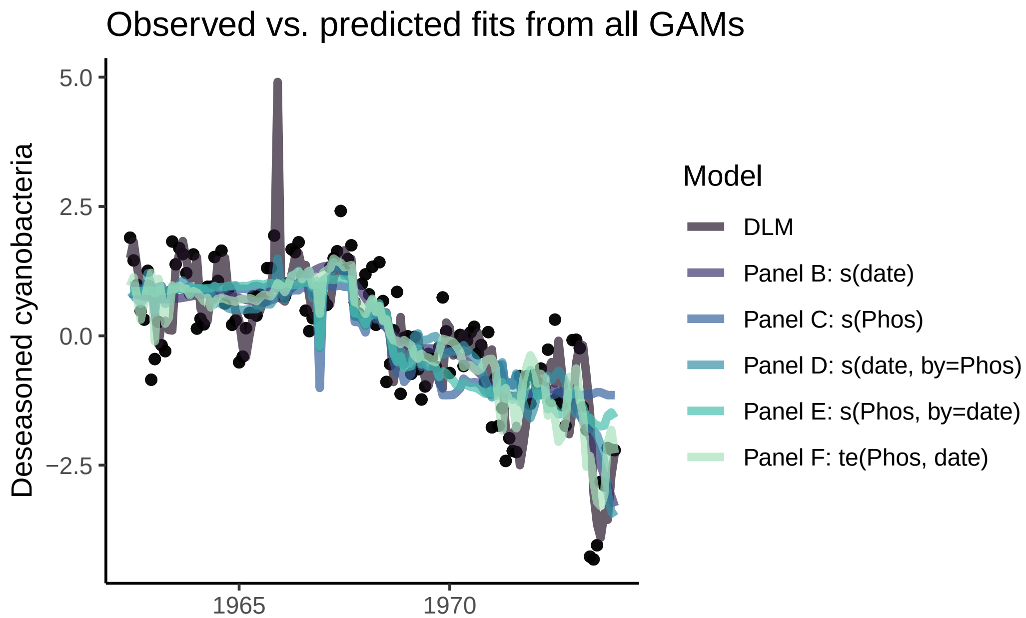

Figure S8: Fit of the GAM models (lines) to observed bluegreen algae (cyanobacteria) abundance (black points). Colors denote different specifications of the smoother functions.

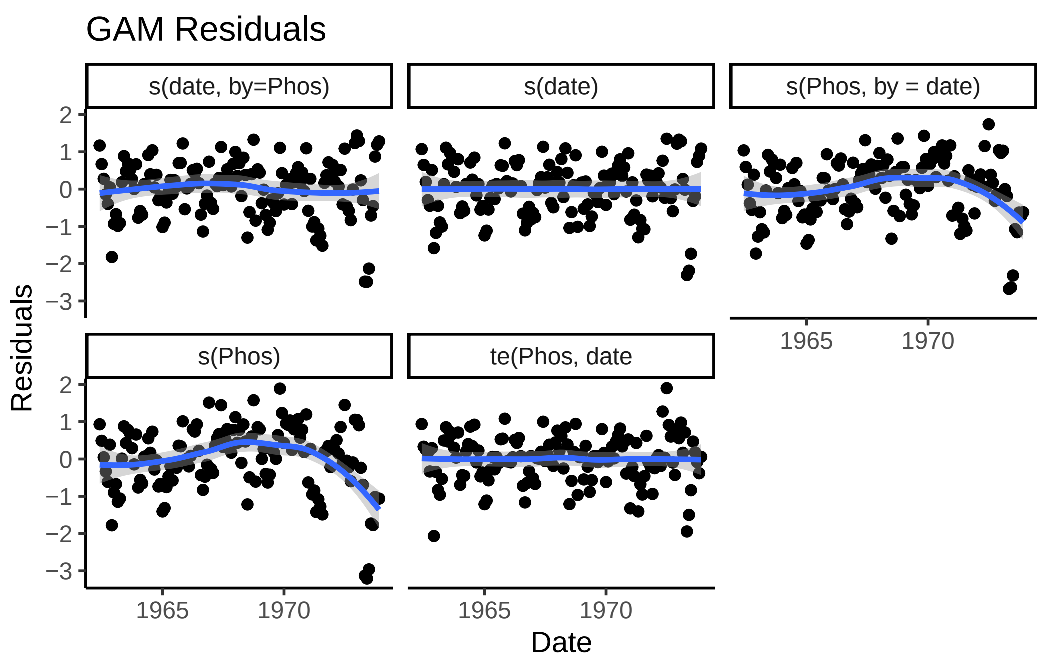

Figure S9: Time series of residuals from the five GAM models where residuals from models that do not have a temporally structured smooth term (s(Phos) and s(Phos, by = date)) have a trend through time.

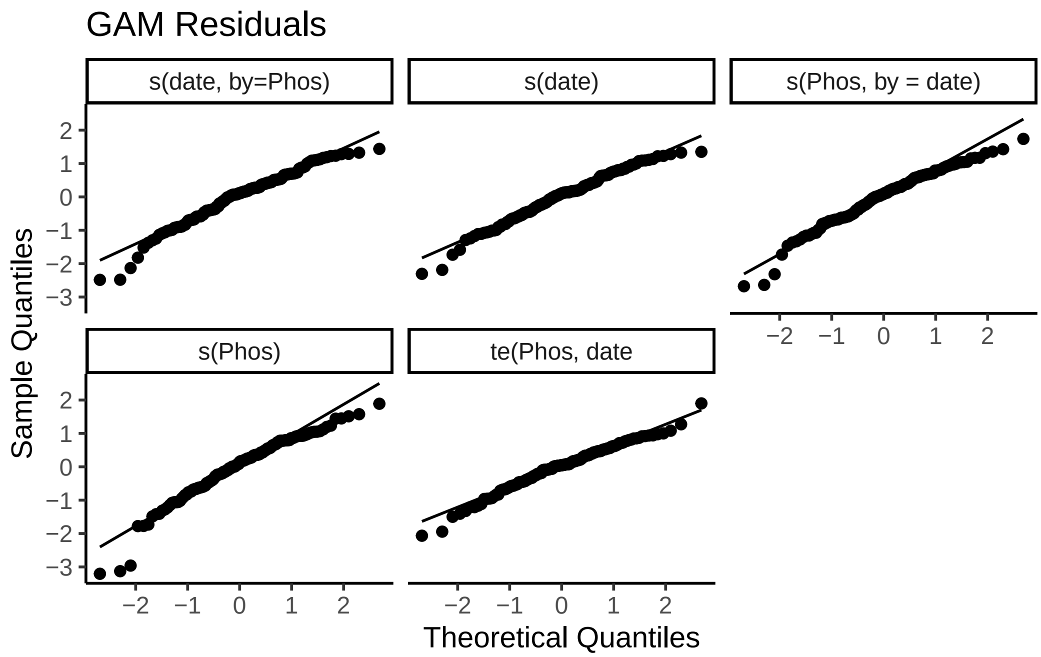

Figure S10: Q-Q for each of the five GAM models where residuals from models that do not have a temporally structured smooth term (s(Phos) and s(Phos, by = date)) have a trend through time.

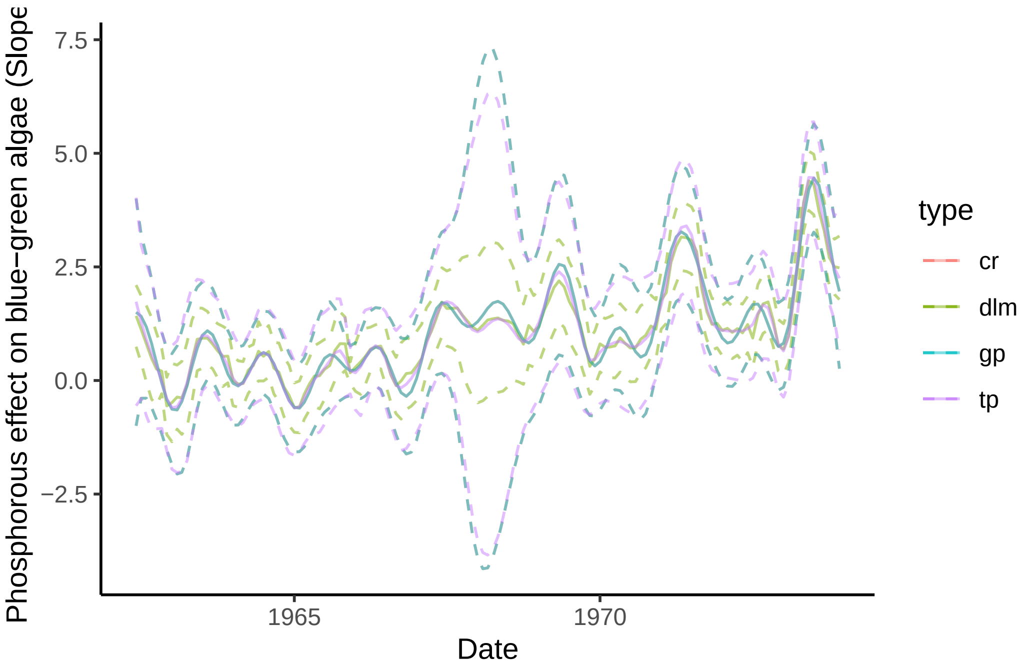

Figure S11: Effect of the choice of smoother function on estimates of time-varying slope parameter where color denotes model type. 3 GAM smoother types (cr, tp, gp) are contrasted with a DLM (dlm).

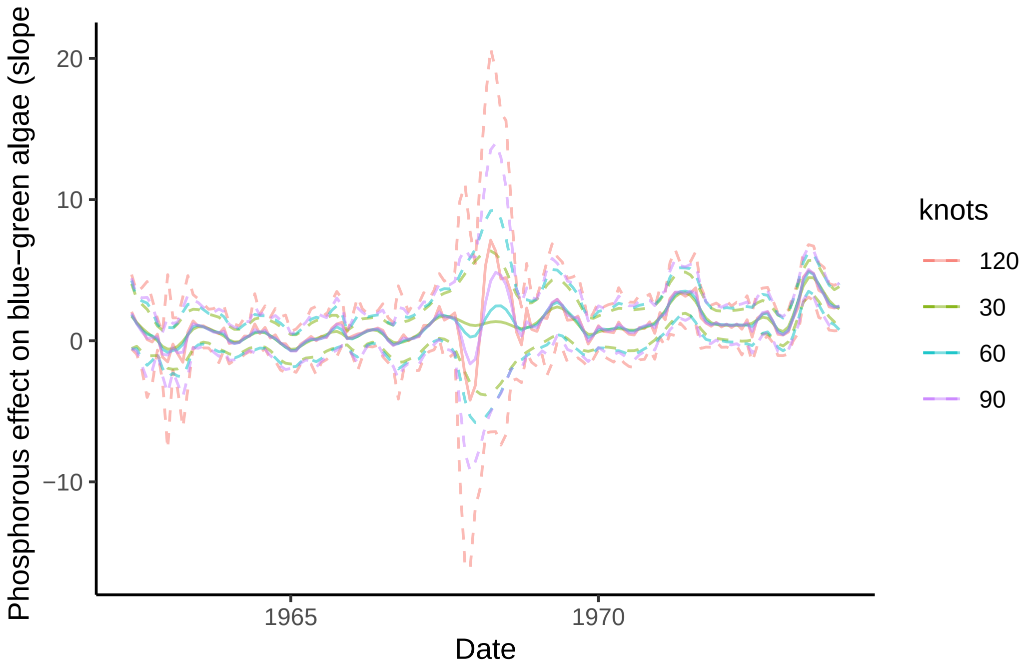

Figure S12: Effect of smoothing degrees of freedom on the standard errors of the estimates of time-varying slope parameter where color denotes model type. We show 4 distinct knots (colors)